TASK ASSIGNMENT & ROADMAP

# Game Developer Intern One-Month Onboarding Plan

Jannah Garden — Unity mobile game, Amal app

| DURATION | PERIOD | PROJECT | ISSUED |
|---|---|---|---|
| 4 weeks | 2 – 27 Nov 2026 | Jannah Garden | 1 Nov 2026 |

## How this month works

Four weeks, each with its own theme and its own build to show. The first two weeks you work in **your own practice project** — you are learning Unity, not touching production. From week three you move into the real **Jannah Garden** repository, starting with audio, which is the safest and most useful place to make a first real contribution.

Every week ends with something that can be opened, installed or listened to. A week with nothing to show is a week we cannot review, so protect the deliverable — if you are running out of time, cut scope and ship the smaller thing.

> **GROUND RULES**
> Ask early — a question costs minutes, a wrong assumption costs days. Share progress mid-week, not only at the deadline. Never commit to the production repo without saying what you are changing first. Read `ONBOARDING.md` in the project root before week three.

## WEEK 1 Environment setup & first build — 2 – 6 Nov

**Goal:** a working Unity installation that can produce a real APK, and your first scene built by hand. By Friday you should be able to install something you made onto a phone.

| # | TASK | WHAT IT MEANS | DONE WHEN |
|---|---|---|---|
| 1 | **Install Unity Hub** | Download Unity Hub from unity.com, install it, and sign in with your Unity account. Hub is the launcher that manages editor versions, licences and projects. | Hub opens, you are signed in, and a personal licence is active. |
| 2 | **Install the editor with Android & iOS SDKs** | Install Unity `6000.3.10f1` — the exact version this project uses. In the install options tick **Android Build Support** (with Android SDK & NDK Tools and OpenJDK) and **iOS Build Support**. It is a large download; start it early. | Both Android and iOS appear as build targets in *File › Build Settings* with no missing-module warnings. |
| 3 | **Decorate a beautiful scene** | Create a new 3D (URP) project and build a small outdoor scene you are proud of — terrain or ground, props, trees, lighting, skybox, a camera angle that flatters it. Use free Asset Store packs. This is a craft exercise: composition and lighting, not scripting. | Scene looks deliberate rather than scattered, runs above 30 FPS in the editor, and you can explain each lighting choice you made. |
| 4 | **Share an APK build** | Switch the platform to Android, set a company and product name, build an APK, install it on a real phone and send it over. Note down any error you hit and how you solved it. | APK shared, installs on a device, and the scene renders correctly on the phone. |

**DELIVERABLE — FRIDAY 6 NOV:** APK of your scene + 3–4 screenshots + a short note listing the setup problems you hit and the fixes.

## WEEK 2 Player control & interaction — 9 – 13 Nov

**Goal:** make that scene playable. A character the player drives, and objects in the world that respond. Same project as week one — keep building on it.

| # | TASK | WHAT IT MEANS | DONE WHEN |
|---|---|---|---|
| 1 | **Third-person controller** | A character that walks, runs and turns, with a camera that follows behind and orbits. Write the movement script yourself rather than dropping in a prefab — you need to understand it. Use the new **Input System**, and add on-screen touch controls (a joystick) since this is a mobile target. Hook up walk/run animations if your model has them. | You can walk the character around the whole scene on a phone without the camera clipping through geometry or the character falling through the ground. |
| 2 | **Object interaction** | Pick at least three objects in the scene and make them interactive: walk close, a prompt appears, tap to act. For example open a door, pick up an item into a simple inventory, and toggle a light. Detect nearby objects with a trigger or a raycast, and keep the interactable behaviour in a small reusable component rather than one script per object. | All three work on device, the prompt appears and disappears at sensible distances, and adding a fourth interactable needs no new script. |
| 3 | **APK build** | Rebuild and share, this time with the controller and interactions in. Check performance on the phone, not just in the editor. | APK shared and playable start to finish by someone who has never seen it. |

**DELIVERABLE — FRIDAY 13 NOV:** Playable APK + a short screen recording showing movement and each interaction + the controller and interaction scripts for review.

## WEEK 3 Unity audio — learn it, then ship it — 16 – 20 Nov

**Goal:** understand Unity's audio stack well enough to own the sound of a real game, and prove it with looping music and a working SFX set. This is the week you start looking at the Jannah Garden repository.

| # | TASK | WHAT IT MEANS | DONE WHEN |
|---|---|---|---|
| 1 | **Learn the audio system** | Audio Source and Audio Listener, 2D versus 3D sound and rolloff, Audio Mixer groups, buses and effects, and the import settings that decide memory cost — Load Type, compression format and quality, Force To Mono, Preload. Learn why a long music track and a short click want completely different settings. Then read `Assets/Scripts/Audio Manager/` in Jannah Garden (`AudioManager`, `GameSfx`, `ButtonClickSfx`) and the **Jannah › Audio** editor menu, so you know how this project actually plays a sound. | You can explain, in your own words, how a button click travels from the tap to the speaker in this project — and which import settings you would pick for music, for a click, and for a looping ambience, with reasons. |

| # | TASK | WHAT IT MEANS | DONE WHEN |
|---|---|---|---|
| 2 | **Implement looping background music** | Add background music that loops seamlessly, survives scene loads without restarting, fades in and out instead of cutting, and respects a volume setting. Route it through a mixer group separate from SFX. A gap or a click at the loop point is a bug — trim the clip until it is gone. | Music loops with no audible seam, keeps playing across a scene change, and the volume control works. Report its memory cost in the Profiler. |
| 3 | **Find and assign SFX** | Source royalty-free sound effects for the common interactions — button press, purchase, reward, pickup, error, footsteps — and wire them in. Use only clearly licensed sources and record the licence for every file. Keep the naming consistent and the import settings correct. Run **Jannah › Audio › Report Unmapped Effects** to see what the project expects. | Every sourced clip is named consistently, correctly imported, mapped to its trigger, and listed in a sheet with its source URL and licence. |

**DELIVERABLE — FRIDAY 20 NOV:** Build with music and SFX working + a licence sheet for every audio file (name, source, licence, where it is used) + your written notes on the audio system.

## WEEK 4 Jannah Garden audio audit & fixes — 23 – 27 Nov

**Goal:** a full pass over the real game's sound, then fix it. This week is half QA, half implementation — and the QA half is shared with the QA team, so compare notes with them.

| # | TASK | WHAT IT MEANS | DONE WHEN |
|---|---|---|---|
| 1 | **Play the game and audit every sound** | Play Jannah Garden properly, start to finish, on a device — onboarding, shop, buying and placing items, treasure boxes, question tasks and Dhikr, XP and area unlocks, photo mode, the minimap, Outer Garden, every menu and popup. For each moment log one row: where you were, what played (or did not), and your verdict — **OK**, **missing**, **wrong**, or **needs change**. Use the sheet overleaf. Be exhaustive; a half audit is worse than none because it reads as complete. | Every screen and interaction in the game has a row. Each problem row says clearly what should happen instead. |
| 2 | **Source replacements & present them** | For every *missing*, *wrong* and *needs change* row, find a replacement from a royalty-free source. Give one or two candidates per slot, not ten. Present them for approval **before** you touch the project — a short list with the context, the candidate clip, the source link and the licence. Nothing goes in unapproved. | Review list submitted, every candidate playable and licence-checked, and you have a decision on each one. |
| 3 | **Assign the approved sounds** | Wire the approved clips in with the right import settings and mixer routing, replacing the wrong ones. Keep the change reviewable — sensible file names, one clear commit, and a note of everything that changed. Then replay the audited flows to confirm each fix. | Approved sounds are in and verified on device, nothing that previously worked has broken, and the audit sheet is updated to show each row resolved. |

**DELIVERABLE — FRIDAY 27 NOV:** Completed audit sheet + the approval list with decisions + a build with the approved sounds in + a one-page summary of what changed.

## Week 4 — audio audit sheet (template)

Copy this into a spreadsheet and fill one row per moment. Add rows freely — the list below is a starting skeleton, not the full scope. Share it with the QA team so you are not auditing the same screens twice.

| # | WHERE (SCREEN / FLOW) | MOMENT / TRIGGER | WHAT PLAYS NOW | VERDICT | WHAT IT SHOULD BE |
|---|---|---|---|---|---|
| 1 | Loading / Init | Loading screen appears | | OK / missing / wrong | |
| 2 | Onboarding | Tutorial step advances | | | |
| 3 | HUD | Any button press | | | |
| 4 | Shop | Panel open / close | | | |
| 5 | Shop | Category tab switch | | | |
| 6 | Shop | Purchase succeeds | | | |
| 7 | Shop | Not enough coins | | | |
| 8 | Placement | Item picked up / dragged | | | |
| 9 | Placement | Placed successfully | | | |
| 10 | Placement | Invalid spot rejected | | | |
| 11 | Treasure box | Box appears / is tapped | | | |
| 12 | Treasure box | Open + reward reveal | | | |
| 13 | Question tasks | Orb found / answer right / wrong | | | |
| 14 | Dhikr | Counter tap / completion | | | |
| 15 | XP | XP gained / level up / area unlock | | | |
| 16 | Player | Footsteps, surface changes | | | |
| 17 | Ambience | Garden background loop, day / night | | | |
| 18 | Photo mode | Enter / shutter / save / share | | | |
| 19 | Outer Garden | Scene entry, fellowship cards | | | |
| 20 | Settings / exit | Volume change, exit confirm | | | |

## How you will be assessed

| AREA | WHAT WE ARE LOOKING FOR |
|---|---|
| **Delivery** | Something real is shared every Friday, and it works on a device — not only in the editor. |
| **Craft** | Scenes look composed, code is readable, assets are named and imported sensibly, performance is checked on a phone. |
| **Judgement** | You ask before assuming, flag problems early, and cut scope rather than miss a deadline silently. |
| **Thoroughness** | The week 4 audit covers the whole game, and every audio file you add has a recorded licence. |
| **Care with the repo** | No unapproved commits, no unrelated files, no broken scenes. You say what you changed. |

### Useful references

- `ONBOARDING.md` — project walkthrough: architecture, scenes, systems, tooling. Read before week 3.
- `Assets/Scripts/Audio Manager/` — `AudioManager`, `GameSfx`, `ButtonClickSfx`.
- **Jannah › Audio** editor menu — populate sound library, re-apply, report unmapped effects.
- `oom_crash_fix_guide.md` — why memory discipline matters here; audio import settings are part of it.
- Unity Manual › Audio, and Unity Learn › Audio for Games, for week 3.